PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Railway Reservation and List management

Student / Team Information

Team Name: *Team #*	
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Anmol Pandey – 2510010778* *anmolpandey2122@gmail.com* 
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Railway reservation systems need to efficiently manage a large number of passengers, limited seats, cancellations, and waiting lists. Managing these operations manually or without an organized data structure can lead to inefficient handling of passenger records and waiting-list allocation.

Our project aims to develop a simplified Railway Reservation and Waiting List Management System using C++. The system will simulate the basic process of booking and cancelling railway tickets. When seats are available, passengers will receive confirmed reservations. When all seats are occupied, new passengers will automatically be added to a waiting list.

The project is mainly motivated by the practical application of Data Structures. A linked list will be used to dynamically manage confirmed passenger records, a queue will manage the waiting list according to the First-In-First-Out (FIFO) principle, and a stack will maintain cancellation history according to the Last-In-First-Out (LIFO) principle.

The system will also demonstrate Object-Oriented Programming concepts through classes for passengers, trains, and the reservation system. The project will help us understand how DSA concepts can be applied to solve a practical problem involving dynamic data and sequential processing.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Modern railway reservation systems are generally computerized and allow passengers to search for trains, check seat availability, book tickets, cancel reservations, and manage waiting lists. When confirmed seats are unavailable, passengers can be placed on a waiting list and may receive confirmation when seats become available due to cancellations.

Large-scale railway systems use sophisticated databases and software infrastructure to handle millions of passenger records and transactions. They also provide features such as online payments, PNR tracking, seat preferences, notifications, and real-time availability.

Our project will not attempt to replicate the complete infrastructure of a real railway reservation platform. Instead, it will provide a simplified simulation focused on the Data Structures involved in reservation management.

The proposed system will demonstrate how Linked Lists, Queues, and Stacks can be used to manage passenger records, waiting-list passengers, and cancellation history respectively.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The main goals of our project are:
** To develop a working railway reservation system using C++.*
** To use a Linked List for storing and managing confirmed passenger records.*
** To use a Queue for maintaining the waiting list.*
** To use a Stack for storing cancellation history.*
** To apply Object-Oriented Programming concepts in the project.*
** To automatically give a cancelled seat to the first passenger in the waiting queue.*
** To provide basic options like booking, cancellation, searching and displaying passenger details.*
** To make the system simple enough to use through a menu-driven interface.*

Milestones

1. *Understand the problem and design the basic system.*
2. *Create the required C++ classes.*
3. *Implement the passenger Linked List.*
4. *Implement the waiting Queue.*
5. *Implement the cancellation Stack.*
6. *Connect all three data structures with the reservation system.*
7. *Add searching and display functions.*
8. *Test the system with different booking and cancellation cases.*
9. *Improve the user interface and prepare the final demonstration.*

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

We will develop the project as a menu-driven C++ application. The user will be provided with different options such as viewing train details, booking a ticket, cancelling a ticket, searching for a passenger, viewing the waiting list and checking cancellation history.

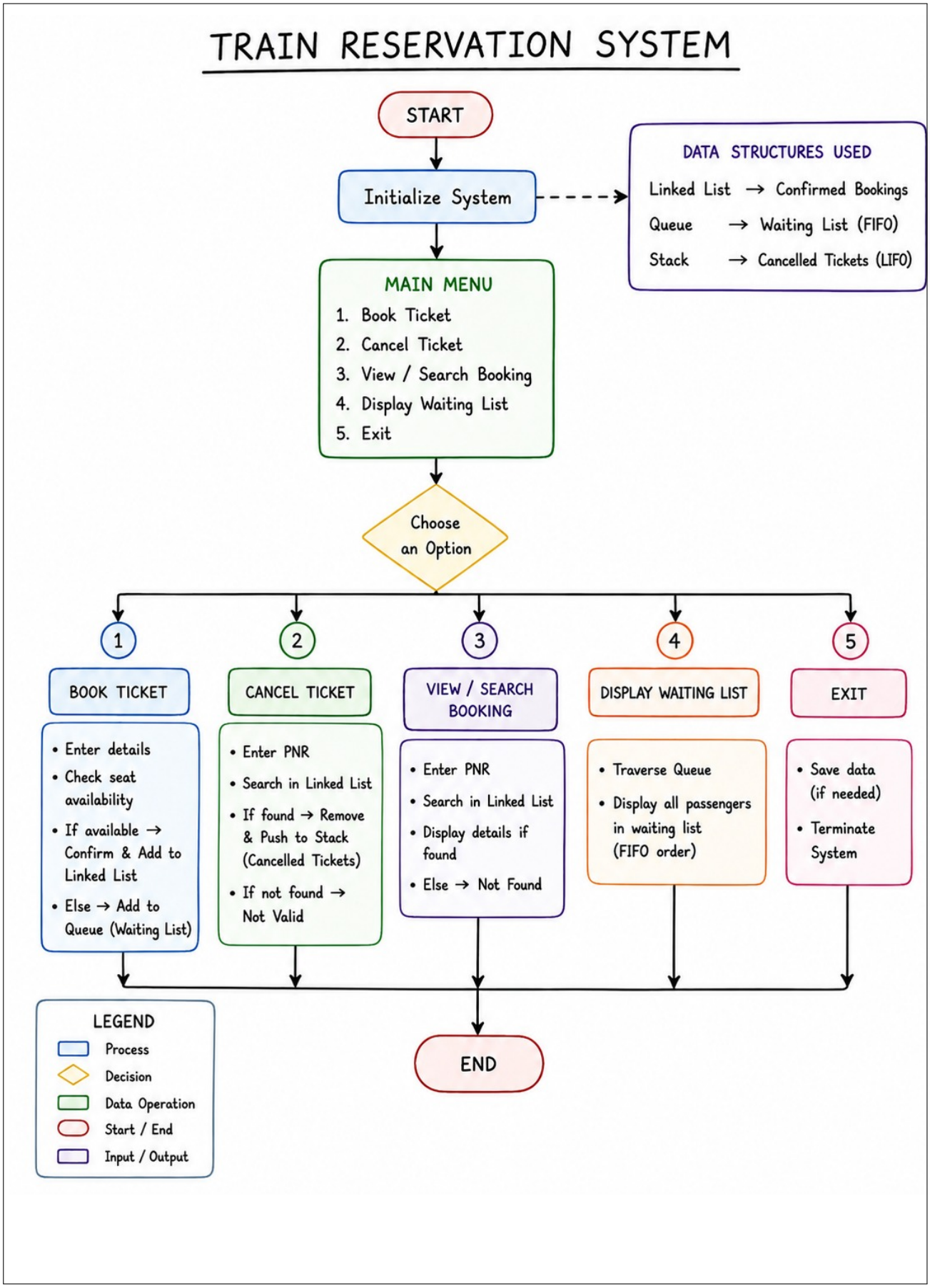
The project will mainly contain classes such as Passenger, Train and ReservationSystem. The Passenger class will store information like passenger name, age, PNR, train number, seat number and booking status.

A Linked List will be used to store confirmed passengers because passenger records can be added or removed during the execution of the program. A Queue will be used for passengers who cannot get a confirmed seat. The first passenger added to the waiting list will be the first one to receive a seat when one becomes available. A Stack will store cancelled ticket information so that the most recent cancellation can be accessed first.

The system will also include automatic promotion of waiting passengers after a cancellation. The project will initially use a console-based interface and standard C++ libraries

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

At the end of the project, we aim to have a working Railway Reservation and Waiting List Management System in C++.

The system will allow users to:

** View available train and seat information.*

** Book a railway ticket.*

** Add passengers to the waiting list when seats are not available.*

** Cancel a confirmed ticket.*

** Automatically promote the first waiting passenger when a seat becomes available.*

** Search for passenger details using PNR or passenger information.*

** Display all confirmed passengers.*

** Display the current waiting list.*

** Display cancellation history.*

The main deliverables will be the C++ source code, project documentation, test cases and a working demonstration. The project will demonstrate the practical use of Linked List, Queue and Stack along with basic OOP concepts.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

*The project will use a limited number of predefined trains. * Each train will have a fixed number of seats.*

** Every passenger will have a unique PNR or passenger ID.*

** The waiting list will follow the FIFO order.*

** Cancellation history will follow the LIFO principle.*

** A cancelled seat will automatically be given to the first passenger in the waiting queue.*

** The project will not include real online payments, live railway data or an actual railway database.*

** The initial version will use a simple console-based interface.*

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. *cppreference – C++ language and standard library reference.*
2. *GeeksforGeeks – Data Structures and Algorithms concepts.*
3. *Programiz – C++ programming and data structure tutorials.*